

Chapter 1 - Sections & Objectives

- 1.1 Personal Computers
 - Explain how to keep personal computer components safe.
 - Explain components in a computer.
 - Explain electrical and ESD safety.
- 1.2 PC Components
 - Explain the features and functions of computer components.
 - Describe cases and power supplies.
 - Describe motherboards.
 - Describe CPUs.
 - Explain types of memory.
 - Describe adapter cards and expansion slots.
 - Describe hard disk drives and SSDs.

Chapter 1 - Sections & Objectives (Cont.)

- 1.2 PC Components (Cont.)
 - Describe optical storage devices.
 - Describe ports, cables and adapters.
 - Describe input devices.
 - Describe output devices.
- 1.3 Disassemble a PC
 - Explain the features and functions of each component in the tool kit.
 - Disassemble a computer.

1.1 Personal Computer

Video Explanation – What is in a computer?

Video Explanation: What is in a Computer?

In this video explanation, you will learn the components that make up a computer:

- Input devices
- Output devices
- Internal components



Personal Computers

Electrical Safety

- Electrical devices have certain power requirements.
- AC adapters are manufactured for specific laptops.
 - Exchanging AC adapters with a different type of laptop or device may cause damage to both the AC adapter and the laptop.
- Some printer parts, such as power supplies, contain high voltage. Check the printer manual for the location of high-voltage components.

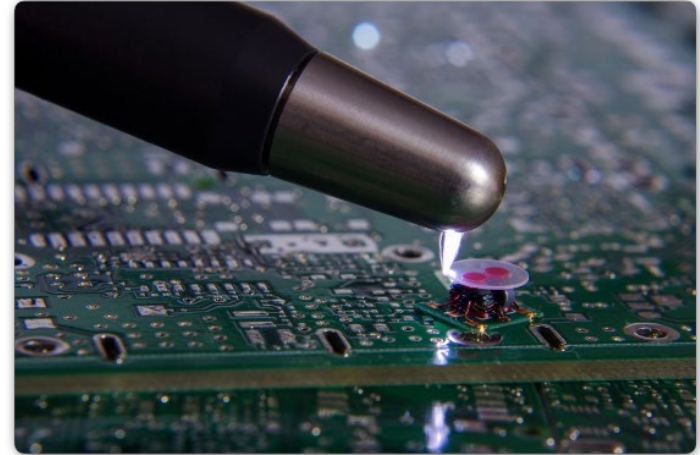


Follow electrical safety guidelines to prevent electrical fires, injuries, and fatalities.

Electrical and ESD Safety

ESD

- Electrostatic discharge (ESD) can occur when there is a buildup of an electric charge that exists on a surface which comes into contact with another differently charged surface.
- ESD can cause damage to computer equipment if not discharged properly.
- At least 3,000 volts of static electricity must build up before a person can feel ESD.
- **Follow these recommendations to help prevent ESD damage:**
 - Keep all components in antistatic bags until you are ready to install them.
 - Use grounded mats on workbenches.
 - Use grounded floor mats in work areas.
 -  Use antistatic wrist straps when working inside computers.



1.2 PC Components

PC Components

Cases

- The case houses the internal components such as the power supply, motherboard, central processing unit (CPU), memory, disk drives, and assorted adapter cards.
- The term **form factor** refers to the physical design and look of a case. Common desktop computers are available in form factors including:

- Horizontal case
- Full-Size Tower
- Compact Tower
- All-in-one

Many case manufacturers may have their own naming conventions, including super tower, full tower, mid tower, mini tower, cube case, and more.



PC Components

Case Types



Mini Tower

MT

Desktop

DT

Small
Form Factor
SFF

Ultra
Form Factor
USFF



PC Components

Power Supplies

- **Converts** alternating-current (AC) wall power into lower voltage direct-current (DC) power
- Computer components use DC.
- Cables, connectors, and components are designed to fit together snugly. Never force any connector or component.
- **CAUTION:** Do not open a power supply.

Electronic capacitors located inside of a power supply can hold a charge for extended periods of time.



PC Components

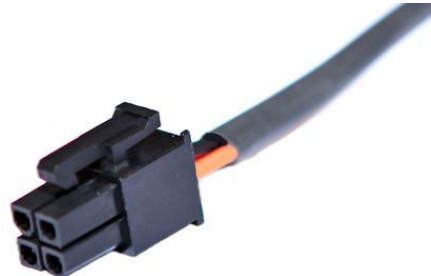
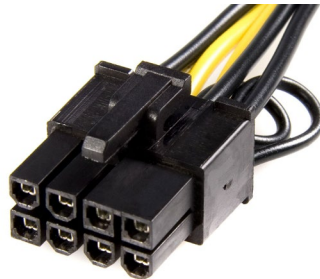
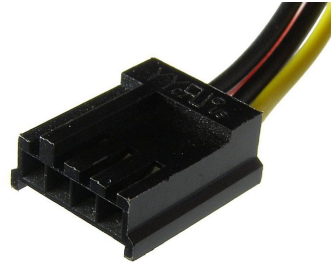
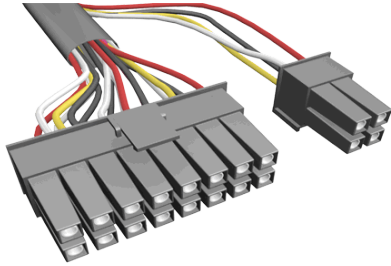
Power Supplies

- Desktop computer power supply form factors include:
 - **Advanced Technology (AT)** – original power supply for legacy computer systems
 - **AT Extended (ATX)** – updated version of the AT
 - **ATX12V** – the most common power supply on the market today
 - **EPS12V** – originally designed for network servers but is now commonly used in high-end desktop models.



PC Components

Power Connectors



- A power supply includes several different connectors. They are used to power various internal components such as the motherboard and disk drives.
- Some examples are:
 - 20-pin or 24-pin slotted connector
 - SATA keyed connector
 - Molex keyed connector
 - Berg keyed connector
 - 4-pin to 8-pin auxiliary power connector
 - 6/8-pin PCIe power connector

Power Supply Voltage

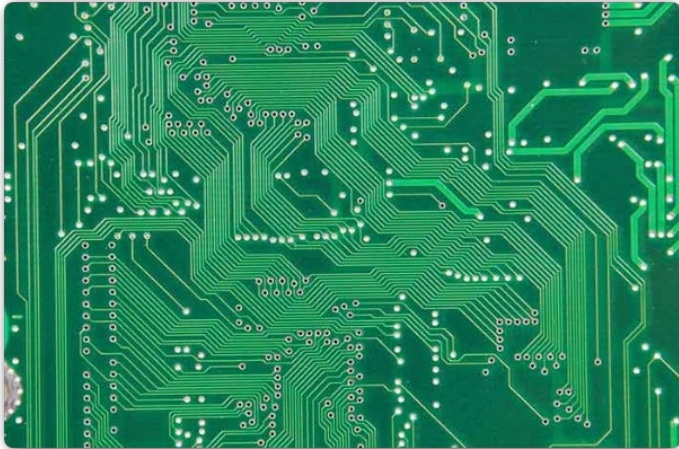
- The different connectors in a power supply also provide different voltages.
- The most common voltages supplied are 3.3 volts, 5 volts, and 12 volts.
- The 3.3 volt and 5 volt supplies are typically used by digital circuits, while the 12 volt supply is used to run motors in disk drives and fans.
- Power supplies can also be single rail, dual rail, or multi rail.
- A rail is the printed circuit board (PCB) inside the power supply to which the external cables are connected.

A computer can tolerate slight fluctuations in power, but a significant deviation can cause the power supply to fail.



PC Components

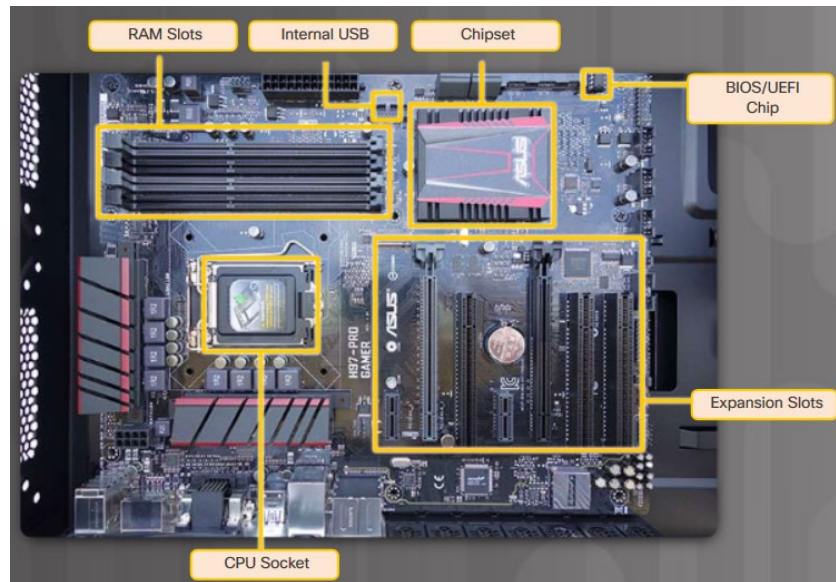
Motherboards



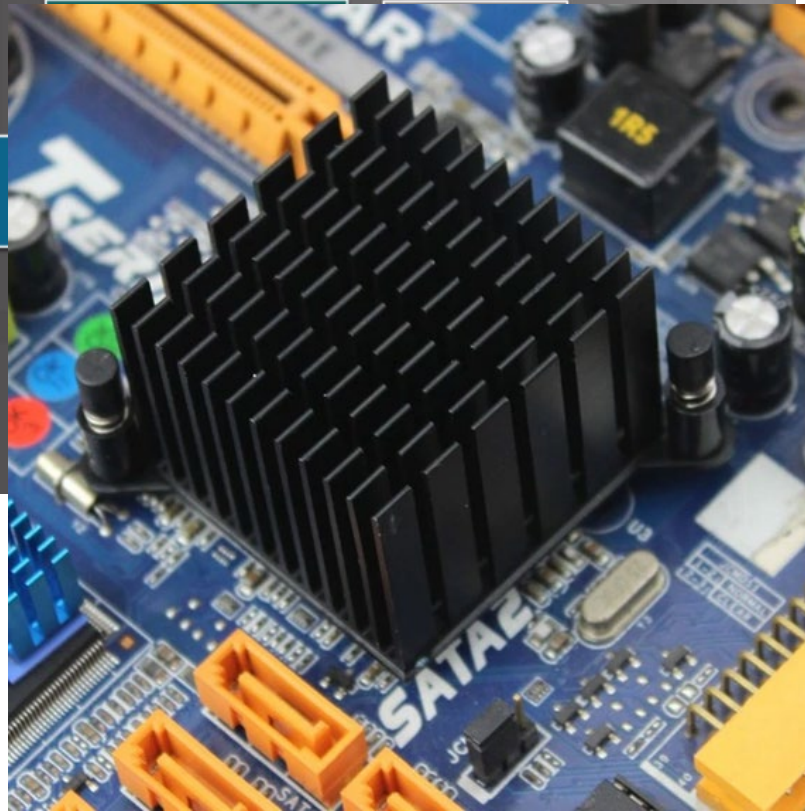
- The motherboard is the backbone of the computer.
- It is a printed circuit board (PCB) that contains buses, or electrical pathways, that interconnect electronic components.
- These components may be soldered directly to the motherboard, or added using sockets, expansion slots, and ports.

Motherboard Components

- Major components on a motherboard include:
 - Central Processing Unit (CPU)
 - Random Access Memory (RAM)
 - Expansion slots
 - Chipset
 - Basic input/output system (BIOS) chip and Unified Extensible Firmware Interface (UEFI) chip
 - SATA connectors
 - Internal USB connector



Motherboard Chipset



- **Chipset** consists of the integrated circuits on the motherboard that control how system hardware interacts with the CPU and motherboard.
- Most chipsets consist of the following two types:
 - **Northbridge** – Controls high speed access to the RAM and video card.
 - **Southbridge** – Allows the CPU to communicate with slower speed devices including hard drives, Universal Serial Bus (USB) ports, and expansion slots.

Motherboard Form Factors

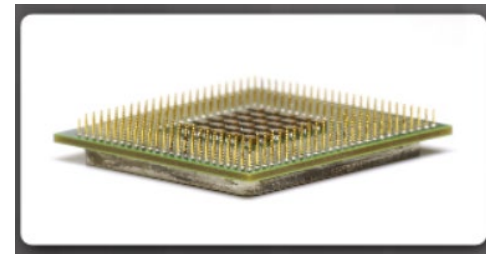
- The form factor of motherboards pertains to the size and shape of the board.
- There are three common motherboard form factors: **Advanced Technology eXtended (ATX)**, **Micro-ATX**, and **ITX**.

Form Factor	Description
ATX	• Advanced Technology eXtended • Most popular form factor • 12 in X 9.6 in (30.5 cm X 24.4 cm)
Micro-ATX	• Smaller footprint than the ATX • Popular in desktop and small form factor computers • 9.6 in X 9.6 in (24.4 cm X 24.4 cm)
Mini-ITX	• Designed for small devices such as thin clients and set-top boxes • 6.7in X 6.7 in (17cm X 17 cm)
ITX	• Comparable form factor to Micro-ATX • 8.5 in X 7.5 in (21.5 cm X 19.1 cm)

The choice of motherboard form factor determines how individual components attach to it, the type of power supply required, and the shape of the computer case.

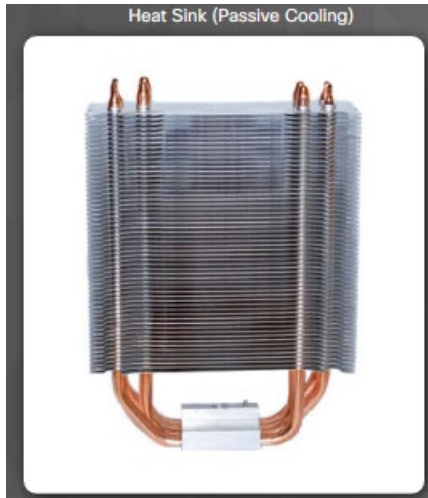
What is CPU?

- The central processing unit (CPU) is responsible for interpreting and executing commands.
- The CPU is a small microchip that resides within a CPU package.
- The CPU socket is the connection between the motherboard and the processor.
- Modern CPU sockets and processor packages are built in following architectures:
 - **Pin Grid Array (PGA)** - the pins are on the underside of the processor package and is inserted into the motherboard CPU socket.
 - **Land Grid Array (LGA)** - the pins are in the socket instead of on the processor.



PC Components

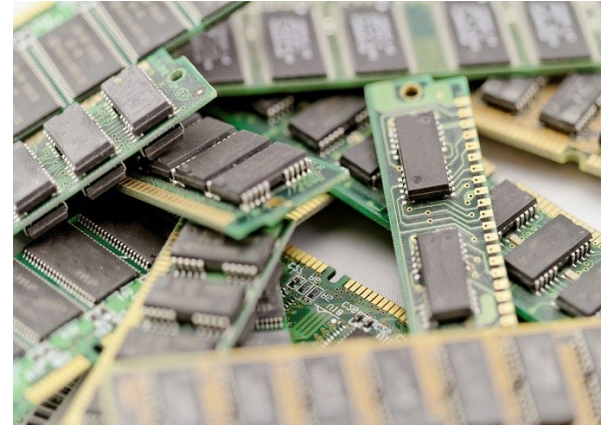
Cooling Systems



- Computer components perform better when kept cool.
- Computers are kept cool using active and passive cooling solutions.
- Active solutions require power while passive solutions do not.
- Passive solutions for cooling usually involve reducing the speed at which a component is operating or adding heat sinks to computer chips.
- A case fan is considered as active cooling.

Types of Memory

- A computer might use different types of memory chips.
- All memory chips store data in the form of bytes.
 - A byte is a block of eight bits stored as either 0 or 1 in the memory chip.
- **Read-Only Memory (ROM)** – such as ROM chip.
- **Random Access Memory (RAM)** is the temporary working storage for data and programs that are being accessed by the CPU. RAM is volatile memory.
- Adding more RAM in a computer enhances the system performance. However, the maximum amount of RAM that can be installed is limited by the motherboard.



PC Components

Types of ROM

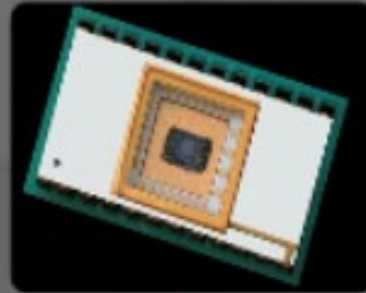
- Types of Read-only Memory (ROM) include:
 - ROM chips.
 - PROM chips.
 - EPROM chips
 - EEPROM chips.



ROM



PROM



EPROM



EEPROM

Types of RAM

- Types of Random Access Memory (RAM) include:
 - Dynamic RAM (DRAM)
 - Static RAM (SRAM)
 - Synchronous Dynamic RAM (SDRAM)
 - Double Data Rate Synchronous Dynamic RAM (DDR SDRAM)
 - DDR2 Synchronous Dynamic RAM (DDR2 SDRAM)
 - DDR3 Synchronous Dynamic RAM (DDR3 SDRAM)
 - DDR4 Synchronous Dynamic RAM (DDR4 SDRAM)
 - GDDR Synchronous Dynamic RAM (GDDR SDRAM)

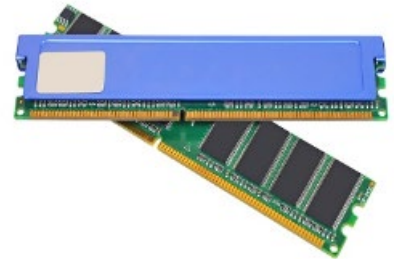
Memory Modules

- Memory chips are soldered to a circuit board to create a **memory module** which is placed into a memory slot on the motherboard.
- Different types of memory modules include: **DIP**, **SIMM**, **DIMM memory**, and **SODIMM**.
- The speed of memory has a direct impact on how much data a processor can process in a given period of time.
- The fastest memory is typically static RAM (**SRAM**) which is used as cache memory



Memory Modules (Cont.)

- The speed of memory has a direct impact on how much data a processor can process in a given period of time.
- The fastest memory is typically static RAM (SRAM) which is used as cache memory for storing the most recently used data and instructions by the CPU.
- The three most common types of cache memory are:
 - **L1 cache** – integrated into the CPU
 - **L2 cache** – was original mounted on the motherboard, but now integrated into the CPU
 - **L3 cache** – used some high-end workstations and server CPUs



Memory Modules (Cont.)

- Memory errors occur when the data is not stored correctly in the memory chips. The computer uses different methods to detect and correct data errors in memory.
- Different types of error checking methods include:
 - **Nonparity** – Nonparity memory does not check errors in memory.
 - **Parity** – Parity memory contains 8 bits for data and 1 bit for error checking.
 - **ECC** – Error Correction Code memory can detect multiple bit errors in memory and correct single bit errors in memory

Common Adapter Cards

- **Adapter cards** increase the functionality of a computer by adding controllers for specific devices or by replacing malfunctioning ports.



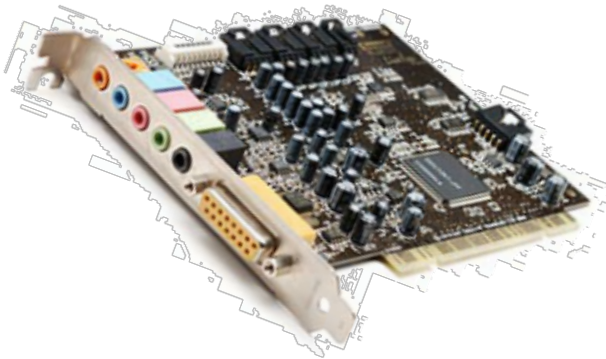
Network Interface Card



Wireless NIC



RAID Card



Sound Card



USB Ports

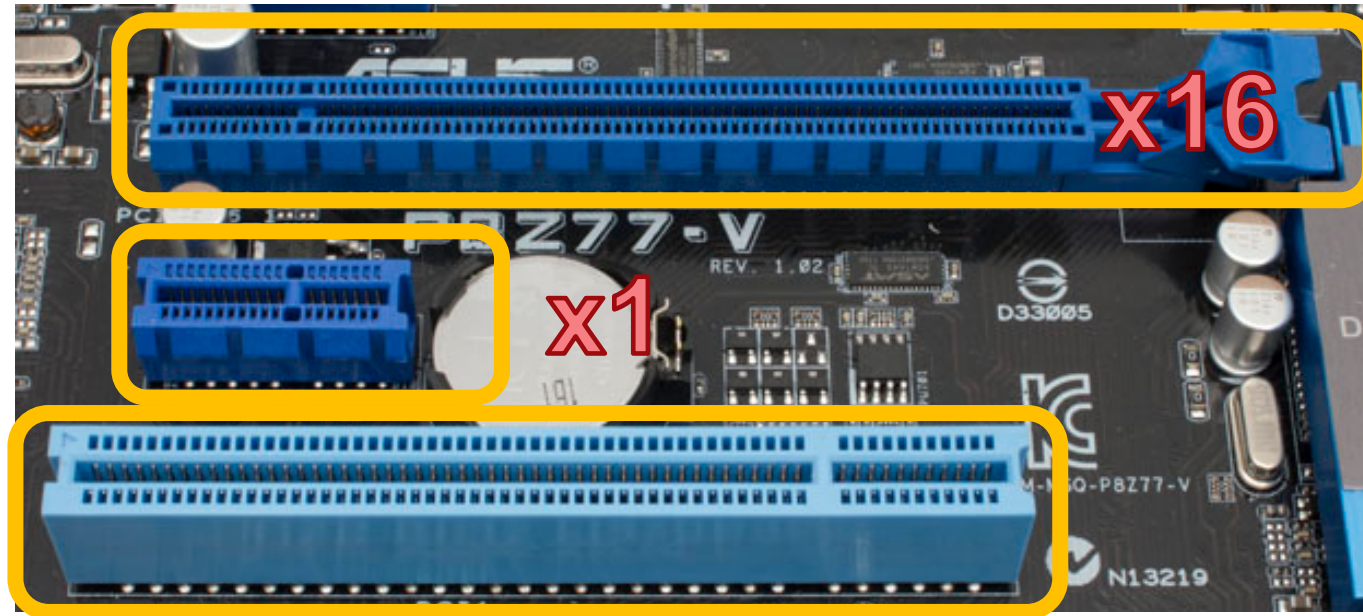


Video Card

PC Components

Expansion Slots

- On the motherboard to install adapter cards.
- Adapter card connector must match the expansion slot.
 - PCI-Express
 - PCI eXtended (PCI-X)
 - Peripheral Component Interconnect (PCI)
 - Riser card
 - Advanced Graphics Port (AGP) (Legacy)



Types of Storage Devices

- Data drives provide **non-volatile** storage of data.
- Some drives have fixed media, and other drives have removable media.
- Data storage devices can be classified according to the media on which the data is stored:
 - **Magnetic** – like hard disk drive and tape drive
 - **Solid state** – like solid state drive
 - **Optical** – like CD and DVD



Hard Disk Drive



Optical Drive



Solid State Drive



Tape Drive

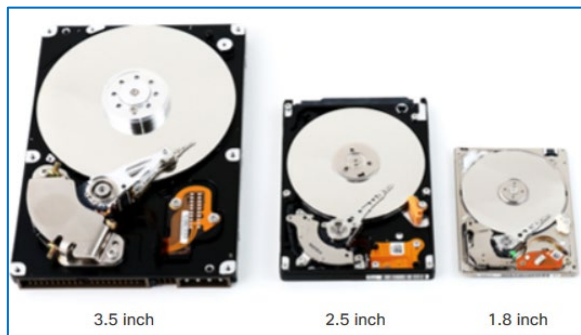
Storage Device Interfaces

- Interior Storage devices **Serial AT Attachment** (SATA) data connections to motherboard
- Legacy interface is **Parallel ATA** (EIDE).
- **Interface standards** define the way that data is transferred, the transfer rates, and physical characteristics of the cables and connectors.
- Three main versions of SATA standard:
 - SATA 1
 - SATA 2
 - SATA 3.
 - Cables & connectors same
 - Data transfer speeds are different.

ATA	Parallel (PATA)	IDE	8.3 Mb/s
		EIDE	16.6 Mb/s
	Serial (SATA)	SATA 1	1.5 Gb/s
		SATA 2	3.0 Gb/s
		SATA 3	6.0 Gb/s

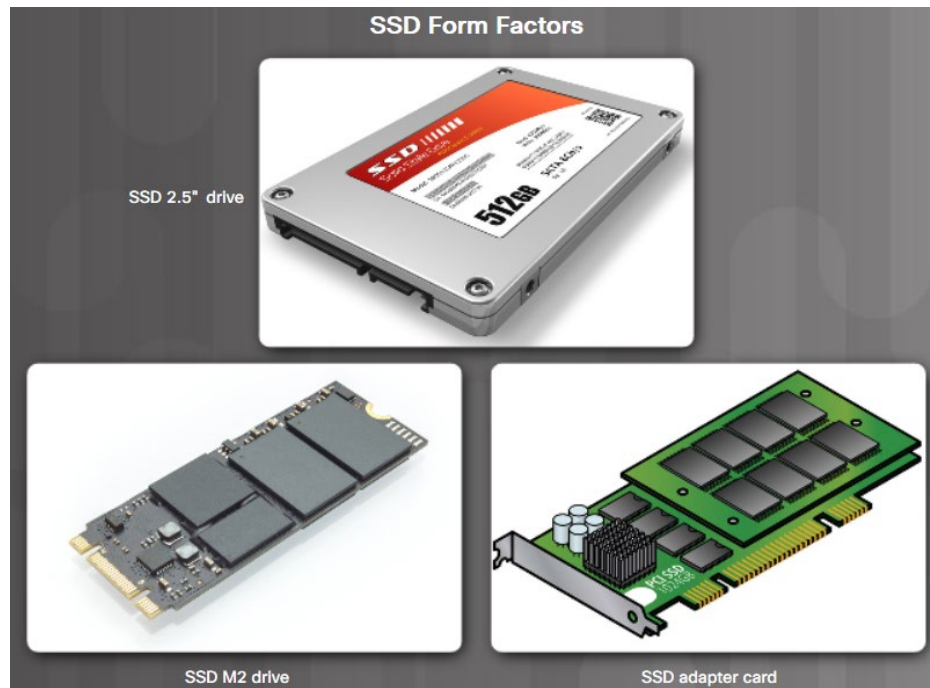
Magnetic Media Storage

- Binary values as magnetized or non-magnetized physical areas of magnetic media.
- Common types of magnetic media storage drives:
 - **Hard Disk Drive** (HDD) – the traditional magnetic disk
 - Storage capacity measured in **gigabytes** (GBs) to **terabytes** (TBs).
 - **Tape Drive** – used for archiving data.
 - Tape drives use a magnetic read/write head and removable tape cartridge



Semiconductor Storage

- **Solid-state drives (SSD)** store data in flash memory.
- SSDs much faster than magnetic HDDs.
 - no moving parts
 - no noise
 - more energy efficient
 - produce less heat
- Three form factors:
 - Disc drive form factor – similar to an HDD
 - Expansion cards – plugs directly into the motherboard
 - **mSata** or **M.2 modules** – these packages may use a special socket.



Semiconductor Storage (Cont.)

- The **Non-Volatile Memory Express (NVMe)** specification was developed specifically to allow computers to take greater advantage of the features of SSDs by providing a standard interface between SSDs, the PCIe bus, and operating systems.
- NVMe allows compliant SSD drives to attach to the PCIe bus without requiring special drivers.
- **Solid State Hybrid Drives (SSHDS)** are a compromise between a magnetic HDD and an SSD.
 - They are faster than an HDD but less expensive than an SSD.
 - They combine a magnetic HDD with onboard flash memory serving as a non-volatile cache.

Optical Storage Devices

- **Optical drives** – removable media storage with lasers to read and write data on optical media.
- Developed because other removable magnetic media such as floppy discs was too small.
- There are three types of optical drives:
 - **Compact Disc (CD)** - audio and data
 - **Digital Versatile Disc (DVD)** - digital video and data
 - **Blu-ray Disc (BD)** - HD digital video and data



Types of Optical Storage Devices (Cont.)

- CD, DVD, and BD media can be pre-recorded (read only), recordable (write once), or re-recordable (read and write multiple times).
- DVD and BD media can also be single layer (SL) or dual layer (DL). Dual layer media roughly doubles the capacity of a single disc.

Optical Media	Description	Storage Capacity
CD-ROM	CD read-only memory media that is pre-recorded	700 MB
CD-R	CD recordable media that can be recorded one time	
CD-RW	CD rewritable media that can be recorded, erased, and re-recorded	
DVD-ROM	DVD read-only memory media that is pre-recorded	4.7 GB (Single-Layer) 8.5 GB (Dual-Layer)
DVD-RAM	DVD rewritable media that can be recorded, erased, and re-recorded	
DVD+/-R	DVD recordable media that can be recorded one time	
DVD+/-RW	DVD rewritable media that can be recorded, erased, and re-recorded	
BD-ROM	Blu-ray read-only media that is pre-recorded with movies, games, or software	25 GB (Single-Layer) 50 GB (Dual-Layer)
BD-R	Blu-ray recordable media that can be recorded one time	
BD-RE	Blu-ray rewritable media that can be recorded, erased, and re-recorded	

Video Ports and Cables



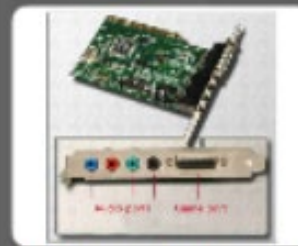
- A video port connects a monitor to a computer using a cable.
- Video ports and monitor cables transfer analog signals, digital signals, or both.
- Video ports and cables include:
 - Digital Visual Interface (DVI)
 - DisplayPort
 - High-Definition Multimedia Interface (HDMI)
 - Thunderbolt 1 or 2
 - Thunderbolt 3
 - Video Graphics Array (VGA)
 - Radio Corporation of America (RCA)

Other Ports and Cables

- Input/output (I/O) ports on a computer connect peripheral devices, such as printers, scanners, and portable drives.
- A computer may have other ports:
 - Personal System 2 (PS/2)
 - Audio and game port
 - Network
 - Serial AT Attachment (SATA)
 - Integrated Drive Electronics (IDE)
 - The Universal Serial Bus (USB)



PS/2



Audio and Gameport



Network



SATA



IDE



USB

Adapters and Converters

- There are many connection standards in use today. These components are called adapters and converters:
 - **Converter** – performing the same function as an adapter but also translates the signals from one technology to the other.
 - **Adapter** – physically connecting one technology to another
- Example of adapters include:
 - DVI to VGA Adapter
 - USB to Ethernet adapter
 - USB to PS/2 adapter
 - DVI to HDMI adapter
 - Molex to SATA adapter
 - HDMI to VGA converter



The Original Input Devices

- Input devices allow the user to communicate with a computer.
- Some of the first input devices include:
 - **Keyboard and Mouse** – these are the two most commonly used input devices
 - **ADF / Flatbed Scanner** – these devices digitize an image or document
 - **Joystick and Gamepad** – these devices are used for playing games
 - **KVM Switch** – a hardware device that can be used to control more than one computer while using a single keyboard, monitor, and mouse



Keyboard and Mouse



ADF / Flatbed Scanner



Joystick and Gamepad



KVM Switch

Newer Input Devices

- Some new input devices include touch screen, stylus, magnetic strip reader, and barcode scanner:
 - **Touch screen** – input devices with touch or pressure sensitive screens
 - **Stylus** – a type of digitizer that allows a designer or artist to create artwork by using a pen-like tool
 - **Magnetic strip reader** – a device that reads information magnetically encoded on the back of plastic cards
 - **Barcode scanner** – a device that reads the information contained in the barcodes affixed to products



Touch screen



Stylus



Magnetic Stripe Reader



Barcode Scanner

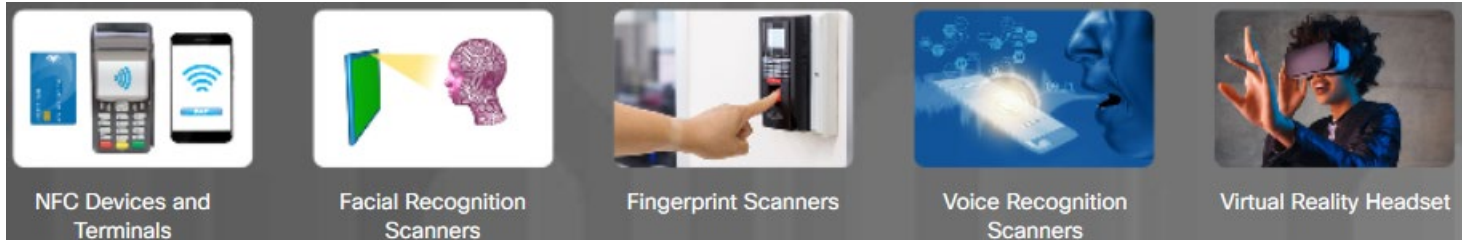
More New Input Devices

- A few newer input devices:
 - **Digital camera** – devices that capture digital images and videos
 - **Webcams** – video cameras that can be integrated into a computer
 - **Signature pad**– a device that electronically captures a person's signature
 - **Smart card reader** – a device used on a computer to authenticate the user. A smart card may be the size of a credit card with an embedded microprocessor that is typically under a gold contact pad on one side of the card.
 - **Microphone** – a device that allows a user to speak into a computer and have their voice digitized



Most Recent Input Devices

- The newest input devices include NFC devices and terminals, facial recognition scanners, fingerprint scanners, voice recognition scanners, and virtual reality headsets:
 - **NFC devices and terminals** – Near Field Communication (NFC) tap to pay devices
 - **Facial recognition scanners** – devices identifying a user based on unique facial features
 - **Fingerprint scanners** – devices identifying a user based on unique fingerprint
 - **Voice recognition scanners** – devices identifying a user based on unique voice
 - **Virtual reality headset** – used with computer games, simulators, and training applications with virtual reality functionalities.



What are Output Devices?

- An output device takes binary information from the computer and converts it into a form that is easily understood by the user.

▪ Examples

- Monitors
- Projectors
- VR headsets
- Printers
- Speakers
- headphones



Monitor



Projector



VR Headset



Printer



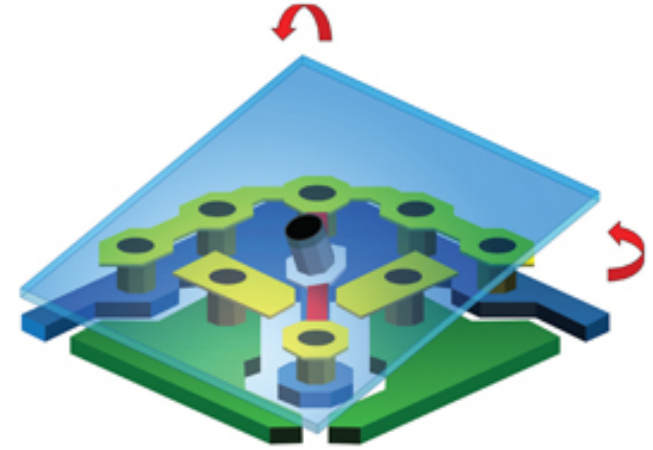
Speakers



Headphones

Monitors and Projectors

- Most monitors use one of three types of technology:
 - Liquid crystal display (**LCD**)
 - Light-emitting diode (**LED**)
 - Organic LED (**OLED**)
- Most video projectors use LCD or **DLP** technology.
 - DLP stands for Digital Light Processing
 - Different projectors have different numbers of lumens, which affects the level of brightness of the projected image.



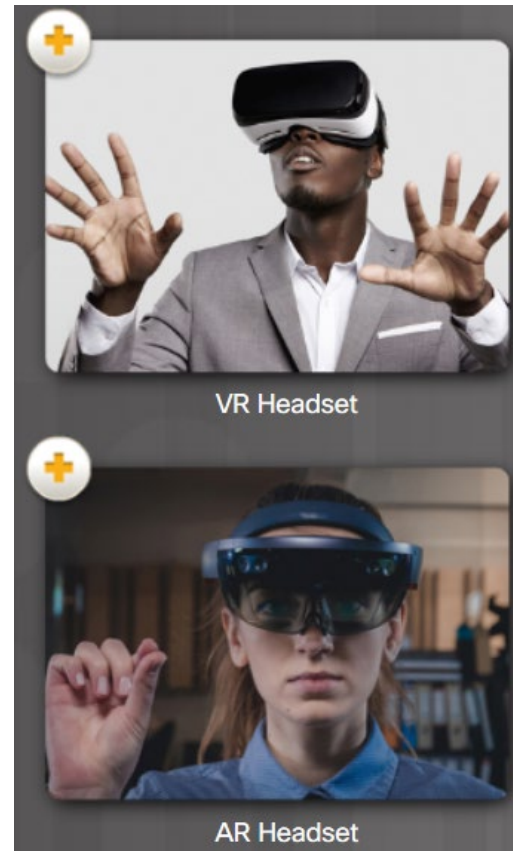
Monitor



Projector

VR and AR Headsets

- **Virtual Reality (VR)** uses computer technology to create a simulated, three-dimensional environment.
- VR headsets encase the upper portion of users' faces which prevents ambient light from entering
- **Augmented Reality (AR)** superimposes images and audio over the real world in real time.
- AR does not close off ambient light to users, allowing them to see their real life surroundings.



PC Components

Printers

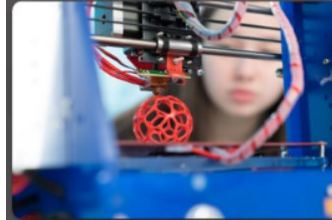
- Printers are output devices that create hard copies of files.
- A **hard copy** might be a on a sheet of paper. It could also be a plastic form created from a 3D printer.
- Different types of printers:
 - **Inkjet, impact, thermal, laser, and 3D** printers
 - Printers use **wired** or **wireless** connections
 - All printers require printing material (such as ink, toner, liquid plastic, etc.)
 - Printers use a driver to communicate with OS



Inkjet Printer



Impact Printer



3D Printer



Thermal Printer

Speakers and Headphones

- **Speakers** are auditory output devices.
- Most computers and mobile devices have audio support
 - Integrated into motherboard
 - On an adapter card.
- **Other auditory devices**
 - **Headphones**
 - Earbuds
 - Earphones
- **Wired or wireless.**
- Some are Wi-Fi or Bluetooth-enabled.



1.3 Computer Disassembly

Video Explanation – Technician's Toolkit

Video Explanation: Technician's Toolkit

In this video explanation, you will learn about the basic tools in a technician's toolkit:

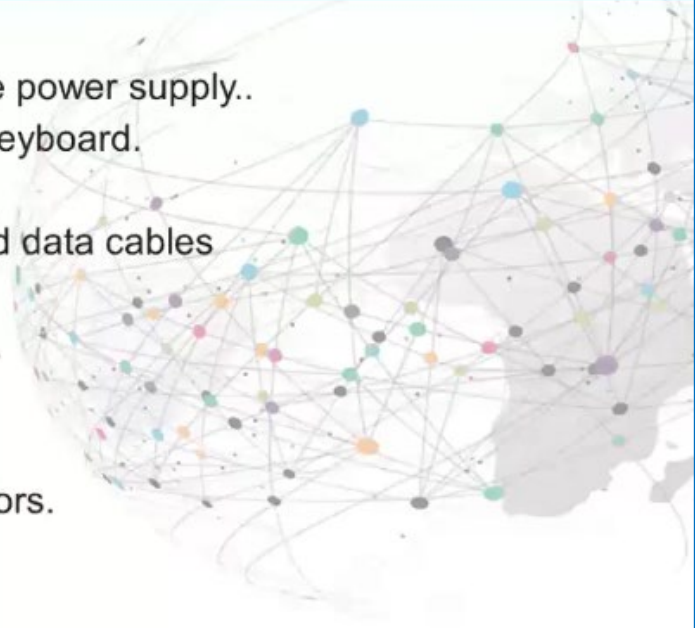
- Anti-static Wrist Strap and Mat
- Compressed Air
- Extraction Tool
- Phillips Head Screwdriver
- Torque Screwdriver
- Cable Tester
- Crimpers



Video Demonstration – Computer Disassembly

Video Demonstration: Disassemble a computer

- **Step 1:** Power off and disconnect the power supply..
- **Step 2:** Disconnect the mouse and keyboard.
- **Step 3:** Remove the case screws.
- **Step 4:** Remove the SATA power and data cables
- **Step 5:** Remove the hard drive.
- **Step 6:** Remove the optical drive.
- **Step 7:** Remove the adapter card.
- **Step 8:** Remove the power supply.
- **Step 9:** Remove front panel connectors.
- **Step 10:** Remove the RAM.



1.4 Chapter Summary

Chapter 1: Introduction to Personal Computer Hardware

- Explain how to keep personal computer components safe.
- Explain the features and functions of computer components.
- Disassemble a computer.